

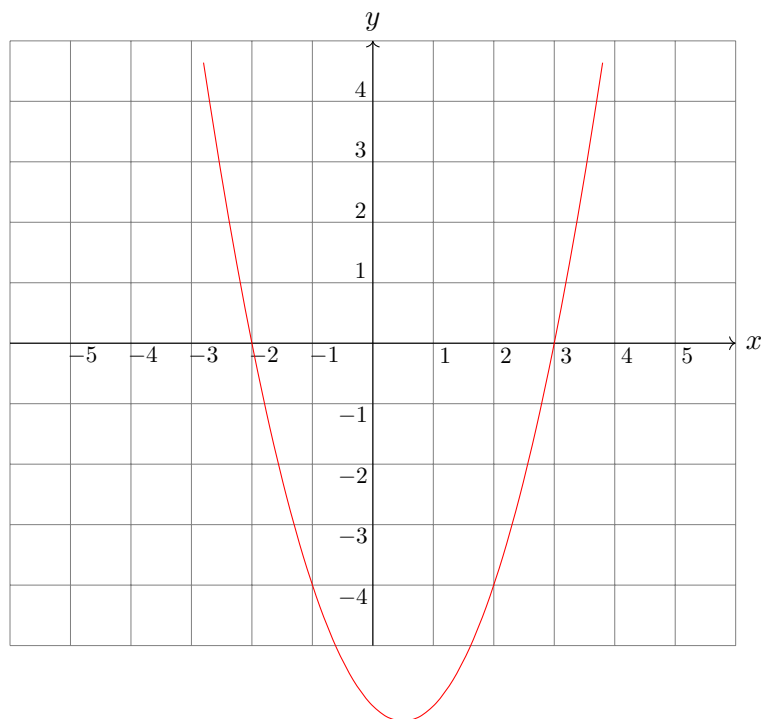
Section 5.3

1. 5 points Evaluate the following integral using the Fundamental Theorem of Calculus. Sketch the graph of the integrand and shade the region whose net area you have found.

$$\int_{-2}^3 (x^2 - x - 6) dx$$

First, note that $F(x) = \frac{1}{3}x^3 - \frac{1}{2}x^2 - 6x$ is an antiderivative of $f(x) = x^2 - x - 6$. As such, $\int_{-2}^3 f(x) dx = F(3) - F(-2)$. Thus,

$$\begin{aligned} \int_{-2}^3 (x^2 - x - 6) dx &= \left(\frac{1}{3}x^3 - \frac{1}{2}x^2 - 6x \right) \Big|_{-2}^3 \\ &= \left(\frac{1}{3}(3)^3 - \frac{1}{2}(3)^2 - 6(3) \right) - \left(\frac{1}{3}(-2)^3 - \frac{1}{2}2(-2)^2 - 6(-2) \right) \\ &= -\frac{125}{6} \approx -20.833 \end{aligned}$$



Section 5.4

2. 5 points Find or approximate all points at which the following function equals its average value on the given interval.

$$f(x) = 8 - 2x \text{ on } [0, 4]$$

$$8 - 2x = \frac{1}{4 - 0} \int_0^4 (8 - 2x) dx$$

$$8 - 2x = \frac{1}{4} (8x - x^2) \Big|_0^4$$

$$8 - 2x = \frac{1}{4} ((8(4) - (4)^2) - (8(0) - (0)^2))$$

$$8 - 2x = \frac{1}{4} (16)$$

$$8 - 2x = 4$$

$$-2x = -4$$

$$x = 2$$